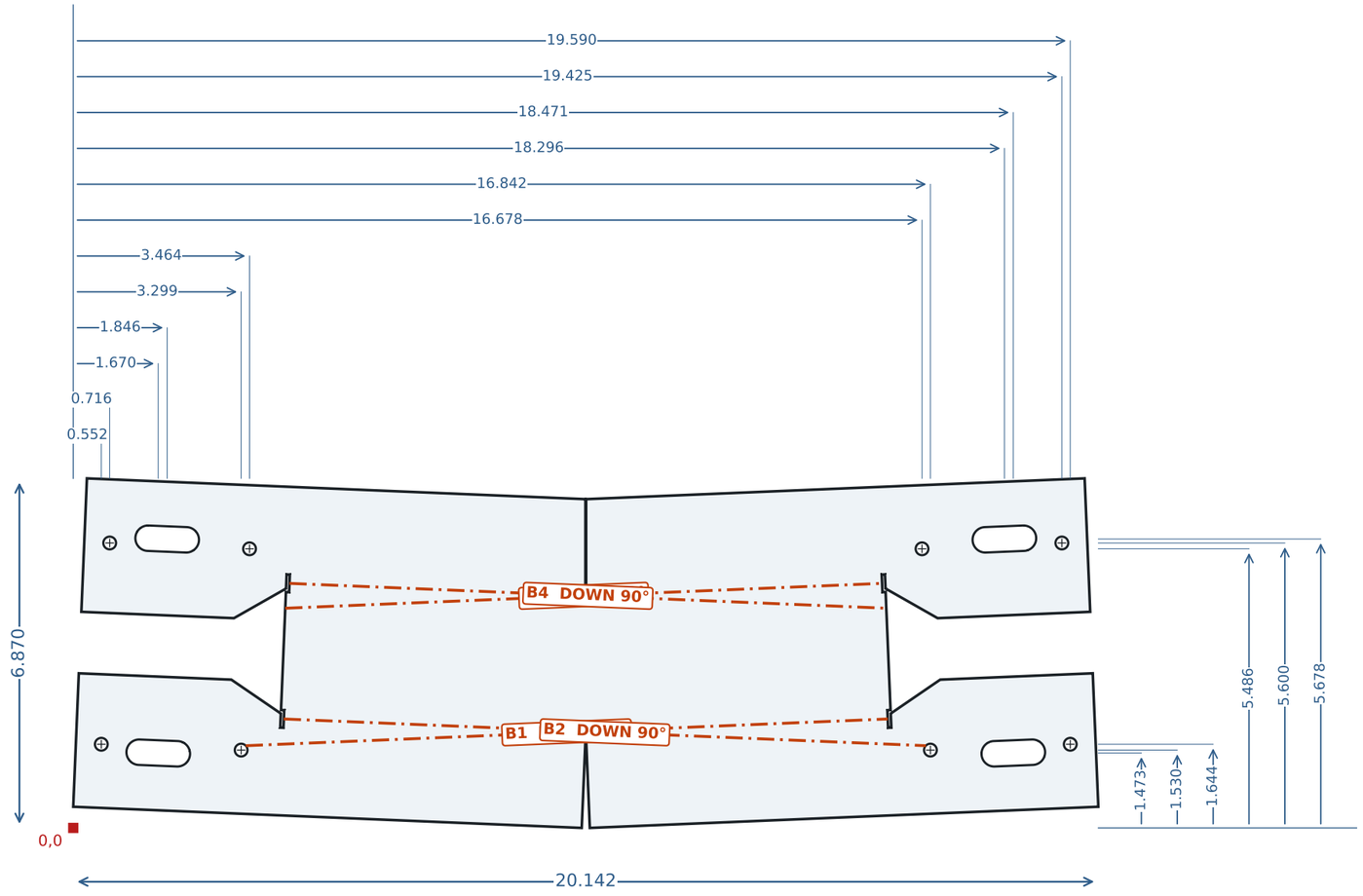
FF-1 · Channel with ears flat pattern, one piece (4 bends 90°, web on top)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 20.142″ × 6.870″ · ≈ 3.89 lb each · QTY 1 per vehicle

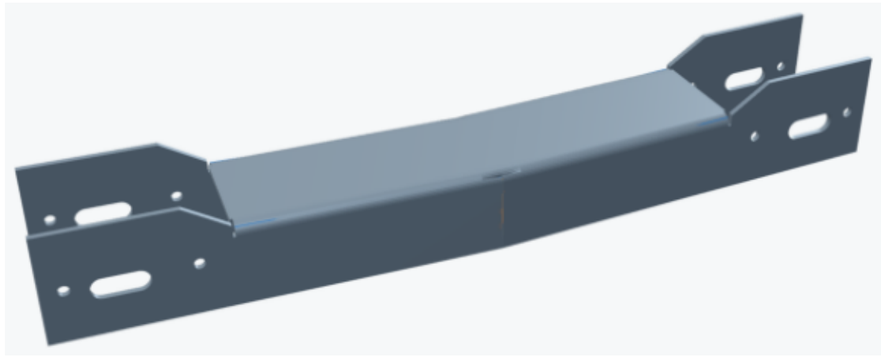
MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FF-1 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)



(shape; thickness now 11 ga 0.120″)

BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	DOWN (away)	R=t 0.120	1	(3.389, 1.616) (16.004, 2.142)
B2	90.0°	DOWN (away)	R=t 0.120	2	(4.137, 2.142) (16.752, 1.616)
B3	90.0°	DOWN (away)	R=t 0.120	3	(4.166, 4.318) (15.893, 4.806)
B4	90.0°	DOWN (away)	R=t 0.120	4	(4.248, 4.806) (15.976, 4.318)

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.250	0.552	1.644	laser
H2	Ø0.250	0.716	5.600	laser
H3	Ø0.250	3.299	1.530	laser
H4	Ø0.250	3.464	5.486	laser
H5	Ø0.250	16.678	5.486	laser
H6	Ø0.250	16.842	1.530	laser
H7	Ø0.250	19.425	5.600	laser
H8	Ø0.250	19.590	1.644	laser
H9	cut-out 1.249×0.531	1.670	1.473	laser
H10	cut-out 1.249×0.531	1.846	5.678	laser
H11	cut-out 1.249×0.531	18.296	5.678	laser
H12	cut-out 1.249×0.531	18.471	1.473	laser

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat.
   Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.

* one blank: the web is a chevron 6"+6" (ends ¼" out); bend the walls 90° down on the dashed lines, each in 2 hits (6" to the middle); outer wall has a V-notch - closes after bending, weld it; inner wall has a slit - opens after bending, weld it; the ears reach past the bend line; 1/16" relief slit at the bend end

FILES

- FF-1_cut.dxf (qty 1 / vehicle)
- FF-1_ref.dxf (reference: + bend lines, material notes)

FF-1 Front channel with ears (chevron 6"+6")	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FF-1 · Channel with ears flat pattern, one piece (4 bends 90°, web on top)	**QTY 1 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 1/1**